An overpartition congruence modulo 7: proof of a conjecture of Wei

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Abstract

Let $\overline{p}(n)$ denote the number of overpartitions of n. Wei conjectured that $\overline{p}(16n) \equiv 0 \pmod 7$ whenever $n \equiv 3 \pmod 8$ and $\left(\frac{n}{7}\right) = 1$. We prove this. Theta-function 2-dissections give an exact expression for $\sum \overline{p}(16n)q^n$; reduction modulo 7 turns it into a holomorphic modular form of weight 18 on $\Gamma_0(224)$ divided by a power series in q^{56}; a twisting argument and Sturm's theorem finish the proof.

1 Introduction

Overpartitions have generating function

$$\sum_{n\geq 0} \overline{p}(n)q^n = \prod_{k\geq 1} \frac{1+q^k}{1-q^k} = \frac{1}{\varphi(-q)},$$

where $\varphi(q) = \sum_{k\in\mathbb{Z}} q^{k^2}$. Wei [5] proved Ramanujan-type congruences $\overline{p}(11(8n+5)) \equiv 0 \pmod{11}$ and $\overline{p}(13\cdot 2^6(8n+7)) \equiv 0 \pmod{13}$ and stated:

Conjecture ([5, Conjecture 4.1]). If $n \equiv 3 \pmod 8$ and $\left(\frac{n}{7}\right) = 1$, then $\overline{p}(2^4 n) \equiv 0 \pmod 7$. Equivalently, $\overline{p}(16(56n+k)) \equiv 0 \pmod 7$ for all $n \geq 0$ and $k \in \{11, 43, 51\}$.

Theorem 1. *For every integer $n \geq 0$ and $k \in \{11, 43, 51\}$,*

$$\overline{p}(16(56n+k)) \equiv 0 \pmod 7.$$

Equivalently, $\overline{p}(16m) \equiv 0 \pmod 7$ for every $m \geq 0$ with $m \equiv 3 \pmod 8$ and $\left(\frac{m}{7}\right) = 1$.

Ryan et al. [3, Theorems 1.3 and 1.5] already give square-prime, Legendre-restricted families, as well as cube-prime families. Our distinction is the actual target: the arguments here are prime to 7, whereas their displayed modulo-7 families contain that prime as a factor. We claim this specialised congruence, not a new general twisting method. Notation: $q = e^{2\pi i z}$, $\operatorname{Im} z > 0$; $E[f](q) = \sum_n f_{2n} q^n$ for $f = \sum f_n q^n$, so $E[f](q^2) = \frac{1}{2}(f(q) + f(-q))$; and $S_k(q) = \sum_n \overline{p}(kn)q^n$. Negative-index coefficients are zero.

2 Dissections

We use three classical identities (Berndt [1, Ch. 16, Entry 25]):

(a) $\varphi(q) + \varphi(-q) = 2\varphi(q^4)$, (b) $\varphi(q)\varphi(-q) = \varphi(-q^2)^2$, (c) $\varphi(q)^2 + \varphi(-q)^2 = 2\varphi(q^2)^2$.

Lemma 2.

$$S_2 = \frac{\varphi(q^2)}{\varphi(-q)^2}, \quad S_4 = \frac{\varphi(q)^3}{\varphi(-q)^4}, \quad S_8 = \frac{E[\varphi^7](q)}{\varphi(-q)^8}, \quad S_{16} = \frac{E\big[E[\varphi^7]\,\varphi^8\big](q)}{\varphi(-q)^{16}}.$$

Proof. By (b), $1/\varphi(-q) = \varphi(q)/\varphi(-q^2)^2$, and $E[\varphi](q) = \varphi(q^2)$ by (a), giving S_2. By (b) again, $S_2 = \varphi(q^2)\varphi(q)^2/\varphi(-q^2)^4$, and $E[\varphi^2] = \varphi^2$ by (c), giving S_4. Writing $S_4 = \varphi(q)^7/\varphi(-q^2)^8$ (by (b)) gives S_8, and writing $1/\varphi(-q)^8 = \varphi(q)^8/\varphi(-q^2)^{16}$ gives S_{16}. □

3 Reduction modulo 7

Reduction modulo 7 is a ring homomorphism $\mathbb{Z}[[q]] \to \mathbb{F}_7[[q]]$ commuting with E and with inversion of series with constant term 1, and $f(q)^7 \equiv f(q^7)$.

Lemma 3. *With $X = \varphi(q)^4$, $Y = \varphi(-q)^4$: $E[\varphi^8] = 8X^2 - 8XY + Y^2$.*

Proof. Put $a = \varphi(t)^2$, $b = \varphi(-t)^2$ with $t^2 = q$. By (c) and (b), $a + b = 2\varphi(q)^2$ and $ab = \varphi(-q)^4$, so $a^2 + b^2 = 4X - 2Y$ and $a^4 + b^4 = (4X - 2Y)^2 - 2Y^2$. Then $E[\varphi^8](q) = \frac{1}{2}(a^4 + b^4)$. □

Since $\varphi^7 \equiv \varphi(q^7)$, $E[\varphi^7] \equiv \varphi(q^{14})$ and

$$E[E[\varphi^7]\varphi^8] \equiv \varphi(q^7)E[\varphi^8] \equiv \varphi(q^7)(X^2 - XY + Y^2) \qquad (\text{as } 8 \equiv 1).$$

Also $1/\varphi(-q)^{16} \equiv \varphi(-q)^5/\varphi(-q^7)^3$, and iterating (b),

$$\frac{1}{\varphi(-q^7)^3} = \frac{\varphi(q^7)^3\varphi(q^{14})^6\varphi(q^{28})^{12}}{\varphi(-q^{56})^{24}}.$$

Multiplying numerator and denominator by $\varphi(q^{56})$:

$$S_{16}(q) \equiv \frac{\widetilde{H}(q)}{G(q^{56})} \pmod 7, \tag{1}$$

where

$$\widetilde{H} = (X^2 - XY + Y^2)\,\varphi(-q)^5\varphi(q^7)^4\varphi(q^{14})^6\varphi(q^{28})^{12}\varphi(q^{56}), \qquad G = \varphi(-q)^{24}\varphi(q).$$

The extra factor $\varphi(q^{56})$ makes the numerator's weight integral while preserving the denominator's support on multiples of 56 and its constant term 1. Write $\widetilde{H} = \sum K(m)q^m$. Since $1/G(q^{56})$ is a series in q^{56}, $\overline{p}(16n) \equiv \sum_{J=0}^{\lfloor n/56 \rfloor} e_J K(n - 56J)$ with $n - 56J \equiv n \pmod{56}$. Hence Theorem 1 follows from:

Proposition 4. *$K(m) \equiv 0 \pmod 7$ for all $m \equiv 3 \pmod 8$ with $\left(\frac{m}{7}\right) = 1$.*

4 Modularity and the Sturm bound

With $\varphi(q^d) = \eta(2d\tau)^5/(\eta(d\tau)^2\eta(4d\tau)^2)$ and $\varphi(-q^d) = \eta(d\tau)^2/\eta(2d\tau)$, each of the three monomials X^2Z, XYZ, Y^2Z of $\widetilde{H}$ (with Z the product of the remaining theta factors) is an eta-quotient of weight 18 and level 224. By the Gordon–Hughes–Newman criterion and Ligozat's cusp-order formula [2, Thms. 1.64, 1.65], each lies in $M_{18}(\Gamma_0(224), \chi)$ with $\chi = \left(\frac{14}{\cdot}\right)$ (we checked $\sum \delta r_\delta \equiv \sum (224/\delta) r_\delta \equiv 0 \pmod{24}$, the character, and nonnegative order at every cusp in exact arithmetic). Hence $\widetilde{H} \in M_{18}(\Gamma_0(224), \chi) \cap \mathbb{Z}[[q]]$.

Lemma 5 (Twisting at a sufficient level)**.** *Let $f = \sum a(m)q^m \in M_k(\Gamma_0(N), \chi)$, $k \in \mathbb{Z}$, let C be the conductor of χ, and let ψ be any Dirichlet character modulo M. Then, for $L = \operatorname{lcm}(N, M^2, MC)$,*

$$f \otimes \psi = \sum \psi(m)a(m)q^m \in M_k(\Gamma_0(L), \chi\psi^2).$$

No minimality of L or novelty of this twisting principle is asserted.

Proof. With $T_j = \left(\begin{smallmatrix} 1 & j/M \\ 0 & 1 \end{smallmatrix}\right)$ and $c_j = M^{-1}\sum_{r \bmod M} \psi(r)\zeta_M^{-rj}$ we have $f \otimes \psi = \sum_{j \bmod M} c_j f|T_j$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(L)$, multiplication gives

$$T_j\gamma T_{-d^2j} = \begin{pmatrix} a + cj/M & b - bcdj/M - cd^2j^2/M^2 \\ c & d - cd^2j/M \end{pmatrix}.$$

Its entries are integral since $M^2 \mid c$, its lower-left entry is divisible by N, and its lower-right entry is d modulo C since $MC \mid c$. Thus $(f|T_j)|\gamma = \chi(d)f|T_{d^2j}$. Reindexing gives $c_{d^{-2}j} = \psi(d)^2 c_j$, proving the law, even for imprimitive ψ. For any rational matrix α of positive determinant, choose $\sigma \in \mathrm{SL}_2(\mathbb{Z})$ with $\sigma\infty = \alpha\infty$. Then $\sigma^{-1}\alpha$ is upper triangular with positive dilation ratio; it preserves nonnegative exponents in a holomorphic cusp expansion. Apply this to each translated cusp matrix $\alpha = T_j\gamma$. □

The indicator of $\{m \equiv 3\ (8),\ (\frac{m}{7}) = 1\}$ is a linear combination of real characters modulo 56:

$$\frac{1}{4}\sum_{\psi \bmod 8} \psi(3)\psi(m) \cdot \frac{1}{2}\big(\chi_0^{(7)}(m) + (\tfrac{m}{7})\big).$$

Here $C = 56$ and $L = \operatorname{lcm}(224, 56^2, 56^2) = 3136$. All twisting characters square to the principal character on units, so

$$g := \sum_{m \equiv 3(8),\, (m/7)=1} K(m)q^m \in M_{18}(\Gamma_0(3136), \chi) \cap \mathbb{Z}[[q]].$$

By Sturm's theorem with character [4, 2], the sufficient inclusive bound is

$$B = \frac{18}{12} \cdot 3136 \cdot \frac{3}{2} \cdot \frac{8}{7} = 8064.$$

Proposition 6 (computation)**.** *For all $m \le 8064$ with $m \equiv 3 \pmod 8$ and $\left(\frac{m}{7}\right) = 1$ (there are 432 such m), $K(m) \equiv 0 \pmod 7$. There are 370 nonzero complementary controls and $K(0) = 1$.*

This proves Proposition 4 and hence Theorem 1.

5 Computations

All proof checks are exact. From the collection root run `python verify.py mod7`. The identity checker compares the dissections with a separate overpartition recurrence; the modularity checker verifies the eta conditions; and both FLINT and a standard-library implementation compute the inclusive bound 8064, obtaining the 432 zero coefficients and 370 nonzero complementary controls of Proposition 6. The original larger-bound calculation is historical corroboration, not required by this proof. These are reproducible computational certificates, not formal proof-assistant verification.

For a compact modularity audit, the exponent vectors of X^2Z, XYZ, Y^2Z, in the order $\delta = (1,2,4,7,14,28,56,112,224)$, are

$$(-6,35,-16,-8,8,-2,46,-19,-2), \quad (10,11,-8,-8,8,-2,46,-19,-2),$$

$$(26,-13,0,-8,8,-2,46,-19,-2).$$

Their orders at cusp denominators $(1,2,4,7,8,14,16,28,32,56,112,224)$ are respectively

$$(70,120,6,10,12,72,2,42,0,84,14,0),$$

$$(126,64,6,18,12,64,2,42,0,84,14,0),$$

$$(182,8,6,26,12,56,2,42,0,84,14,0).$$

These are width-normalised cusp orders, so their nonnegativity proves holomorphy at every cusp.

Collection and verification

This paper is part of the revised collection, version 1.1.0-candidate, DOI `https://doi.org/10.5281/zenodo.23319157`. Companion papers and exact verification code are archived together. The statements are unrefereed computer-assisted proof candidates; no formal verification or unaffiliated reproduction is asserted.

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